

Ultrashort Pulse Propagation Equation

Full derivation in SI units

This document rederives the generalized nonlinear envelope equation entirely in SI units. It follows the same logical path as the Gaussian-unit derivation, but every prefactor is recomputed rather than translated, so the two can be compared independently. Section 6 on self-steepening is written to be self-contained: it can be edited or lifted out without touching the rest.

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1 Conventions

Envelope. The real electric field is written as a carrier at ω_0 modulated by a slowly varying complex envelope $\mathcal{E}$:

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{2} [\mathcal{E}(\mathbf{r}, t) e^{i\psi} + \text{c.c.}], \quad \psi \equiv k_0 z - \omega_0 t. \quad (1)$$

With this convention the peak of the real field is $|\mathcal{E}|$, the cycle average of its square is $\langle \mathbf{E}^2 \rangle = \frac{1}{2}|\mathcal{E}|^2$, and the intensity is

$$I = \frac{1}{2} \varepsilon_0 n_0 c |\mathcal{E}|^2 \quad (2)$$

This is the convention used in the solver, where the field array plays the role of $\mathcal{E}$ and the intensity is formed as `0.5*eps0*n0*c*abs(E)**2`. Fixing it once here removes the factor-of-two ambiguities that plague envelope derivations.

Fourier transform. We use

$$\mathbf{E}(\mathbf{r}, t) = \frac{1}{2\pi} \int \mathbf{E}(\mathbf{r}, \omega) e^{-i\omega t} d\omega, \quad (3)$$

so that $\partial_t \rightarrow -i\omega$ and, for the detuning $\Omega = \omega - \omega_0$,

$$\Omega \longleftrightarrow i \frac{\partial}{\partial t}. \quad (4)$$

Every operator below follows from this single rule; getting its sign wrong flips the sign of both dispersion and self-steepening.

Maxwell (SI). For a non-magnetic medium ($\mathbf{B} = \mu_0 \mathbf{H}$),

$$\nabla \cdot \mathbf{D} = \rho_f, \quad \nabla \cdot \mathbf{B} = 0, \quad (5)$$

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \quad \nabla \times \mathbf{H} = \frac{\partial \mathbf{D}}{\partial t} + \mathbf{J}_{\text{ion}} + \mathbf{J}_{\text{pl}} + \mathbf{J}_{\text{STE}}, \quad (6)$$

with $\mathbf{D} = \varepsilon_0 \mathbf{E} + \mathbf{P}$ and $\mu_0 \varepsilon_0 = 1/c^2$.

Check. In the Gaussian document, Ampère's law is written $\nabla \times \tilde{\mathbf{H}} = \nabla \times \tilde{\mathbf{B}}$. That equality holds only because $\mu = 1$ in Gaussian units; it is worth stating explicitly, since it is the step that assumes a non-magnetic medium. Also, $\mathbf{J}_{\text{STE}}$ appears in the wave equation there but is absent from Ampère's law – it should be introduced in both places.

2 From Maxwell to the wave equation

Taking the curl of Faraday's law and using $\nabla \times (\nabla \times \mathbf{A}) = \nabla(\nabla \cdot \mathbf{A}) - \nabla^2 \mathbf{A}$ with $\nabla \cdot \mathbf{E} \simeq 0$ (homogeneous medium):

$$-\nabla^2 \mathbf{E} = -\mu_0 \frac{\partial}{\partial t} \left[\frac{\partial \mathbf{D}}{\partial t} + \mathbf{J} \right], \quad \mathbf{J} \equiv \mathbf{J}_{\text{ion}} + \mathbf{J}_{\text{pl}} + \mathbf{J}_{\text{STE}}.$$

Splitting the polarization into linear and nonlinear parts, $\mathbf{D} = \mathbf{D}^{(1)} + \mathbf{P}_{\text{NL}}$ with $\mathbf{D}^{(1)} = \varepsilon_0 \mathbf{E} + \mathbf{P}^{(1)}$:

$$\nabla^2 \mathbf{E} - \mu_0 \frac{\partial^2 \mathbf{D}^{(1)}}{\partial t^2} = \mu_0 \frac{\partial^2 \mathbf{P}_{\text{NL}}}{\partial t^2} + \mu_0 \frac{\partial \mathbf{J}}{\partial t}. \quad (7)$$

Comparing with the Gaussian form, the entire unit conversion is the single substitution

$$\frac{4\pi}{c^2} \longrightarrow \mu_0 = \frac{1}{\varepsilon_0 c^2} \quad \text{i.e.} \quad 4\pi \longrightarrow \frac{1}{\varepsilon_0}. \quad (8)$$

In the frequency domain, with $\mathbf{D}^{(1)}(\omega) = \varepsilon_0 \varepsilon_r(\omega) \mathbf{E}(\omega)$,

$$\nabla^2 \mathbf{E} + \frac{\omega^2}{c^2} \varepsilon_r(\omega) \mathbf{E} = -\mu_0 \omega^2 \underbrace{\left[\mathbf{P}_{\text{NL}} + \frac{i\mathbf{J}}{\omega} \right]}_{\equiv \mathcal{S}(\mathbf{r}, \omega)}. \quad (9)$$

The grouping $\mathcal{S}$ plays the role of the symbol ka in the Gaussian document.

3 Envelope and dispersion

Inserting $\mathbf{E}(\mathbf{r}, \omega) \simeq \mathcal{E}(\mathbf{r}, \Omega) e^{ik_0 z}$ and expanding the Laplacian,

$$\left[\nabla_{\perp}^2 + \partial_z^2 + 2ik_0 \partial_z - k_0^2 \right] \mathcal{E} + k^2(\omega) \mathcal{E} = -\mu_0 \omega^2 \mathcal{S} e^{-ik_0 z},$$

with $k^2(\omega) = \varepsilon_r(\omega) \omega^2 / c^2$. Expanding $k(\omega)$ around ω_0 ,

$$k(\omega) = k_0 + k_1 \Omega + \underbrace{\sum_{n \geq 2} \frac{k_n}{n!} \Omega^n}_{\equiv L}, \quad k_1 = \frac{1}{v_g} = \frac{n_g}{c}, \quad (10)$$

so that $k^2 - k_0^2 = 2k_0 k_1 \Omega + k_1^2 \Omega^2 + 2k_0 L + 2k_1 L \Omega + L^2$. Applying rule (4) and moving to the retarded frame $\tau = t - k_1 z$, $z' = z$ (which removes the $2k_0 k_1 \Omega$ term), and dropping ∂_z^2 (SVEA):

$$\left[\nabla_{\perp}^2 + 2ik_0 \partial_{z'} \left(1 + \frac{ik_1}{k_0} \partial_{\tau} \right) + 2k_0 \hat{L} \left(1 + \frac{ik_1}{k_0} \partial_{\tau} \right) + \hat{L}^2 \right] \mathcal{E} = -\mu_0 \omega_0^2 \hat{T}^2 \mathcal{P}_{\text{NL}} + \dots \quad (11)$$

where $\hat{L} = \sum_{n \geq 2} \frac{k_n}{n!} (i \partial_{\tau})^n$.

4 The two operators $\hat{U}$ and $\hat{T}$

Two distinct first-order temporal operators appear, and they are easy to conflate because they look alike.

Space-time focusing, $\hat{U}$. The bracket generated by the retarded-frame change of variables in (11) is

$$\hat{U} = 1 + \frac{k_1}{k_0} \Omega = 1 + \frac{n_g}{n_0} \frac{\Omega}{\omega_0} = 1 + i \frac{n_g}{n_0 \omega_0} \frac{\partial}{\partial \tau}, \quad (12)$$

using $k_1/k_0 = (n_g/c)(c/n_0 \omega_0) = n_g/(n_0 \omega_0)$. It multiplies $\partial_{z'}$ and the dispersion operator; physically it says that different frequency components diffract differently, which couples the transverse and temporal dynamics.

Self-steepening, $\hat{T}$. This one has a completely different origin: it is what the *second time derivative* of the nonlinear polarization produces. With $\mathbf{P}_{\text{NL}} = \frac{1}{2} \mathcal{P}_{\text{NL}} e^{i\psi} + \text{c.c.}$,

$$\begin{aligned} \frac{\partial^2}{\partial t^2} \left[\mathcal{P}_{\text{NL}} e^{-i\omega_0 t} \right] &= \left[\ddot{\mathcal{P}} - 2i\omega_0 \dot{\mathcal{P}} - \omega_0^2 \mathcal{P} \right] e^{-i\omega_0 t} \\ &= -\omega_0^2 \left[1 + \frac{2i}{\omega_0} \partial_t - \frac{1}{\omega_0^2} \partial_t^2 \right] \mathcal{P} e^{-i\omega_0 t} = -\omega_0^2 \hat{T}^2 \mathcal{P} e^{-i\omega_0 t}, \end{aligned} \quad (13)$$

so the square is exact, not an approximation:

$$\hat{T} = 1 + \frac{i}{\omega_0} \frac{\partial}{\partial \tau}. \quad (14)$$

Why the literature writes a single $\hat{T}$. Couairon and Mysyrowicz write the Kerr term with one power of $\hat{T}$, not two. The two statements are consistent. Dividing the propagation equation through by $\hat{U}$ leaves the Kerr term proportional to $\hat{T}^2/\hat{U}$, and comparing (12) with (14),

$$\hat{U} = 1 + i \frac{n_g}{n_0} \frac{1}{\omega_0} \partial_{\tau}, \quad \hat{T} = 1 + i \frac{1}{\omega_0} \partial_{\tau} \quad \implies \quad \hat{U} \simeq \hat{T} \quad \text{when } n_g \simeq n_0. \quad (15)$$

For fused silica at 800 nm, $n_0 = 1.4533$ and $n_g \simeq 1.4671$, so $n_g/n_0 = 1.0095$: the two operators agree to within 1%. Hence

$$\frac{\hat{T}^2}{\hat{U}} \simeq \hat{T}, \quad (16)$$

and the single- $\hat{T}$ form of the literature is recovered. This is the cleanest way to reconcile the two conventions, and it matters for Section 6: the physically meaningful shock coefficient is the one obtained *after* dividing by $\hat{U}$.

5 Nonlinear polarization and n_2

In SI, $\mathbf{P} = \varepsilon_0 \chi^{(1)} \mathbf{E} + \varepsilon_0 \chi^{(3)} \mathbf{E}^3$. Cubing (1),

$$\mathbf{E}^3 = \frac{1}{8} \mathcal{E}^3 e^{3i\psi} + \frac{3}{8} |\mathcal{E}|^2 \mathcal{E} e^{i\psi} + \text{c.c.} \quad (17)$$

The $3\omega_0$ term drives third-harmonic generation and is dropped. Writing $\mathbf{P}_{\text{NL}} = \frac{1}{2} \mathcal{P}_{\text{NL}} e^{i\psi} + \text{c.c.}$,

$$\mathcal{P}_{\text{NL}} = \frac{3}{4} \varepsilon_0 \chi^{(3)} |\mathcal{E}|^2 \mathcal{E}. \quad (18)$$

The refractive index follows from $\varepsilon_r = n_0^2 + \frac{3}{4} \chi^{(3)} |\mathcal{E}|^2$:

$$n = \sqrt{\varepsilon_r} \simeq n_0 + \frac{3\chi^{(3)}}{8n_0} |\mathcal{E}|^2 \equiv n_0 + n_2 I, \quad (19)$$

and inserting I from (2),

$$\boxed{n_2 = \frac{3\chi^{(3)}}{4\varepsilon_0 n_0^2 c} \iff \chi^{(3)} = \frac{4}{3} \varepsilon_0 c n_0^2 n_2} \quad (20)$$

which is the relation already used in the main report. Note that $\chi^{(3)}$ is *not* numerically the same quantity as in Gaussian units: the two differ both by ε_0 (from $\mathbf{P} = \varepsilon_0 \chi^{(3)} \mathbf{E}^3$ versus $\hat{\mathbf{P}} = \chi^{(3)} \hat{\mathbf{E}}^3$) and by the 4π in $\mathbf{D}$. Only n_2 , being defined through the measurable $\Delta n = n_2 I$, is convention-independent.

Kerr source term. Substituting (18) and (20) into the right-hand side of (11), then dividing by $2ik_0$:

$$\begin{aligned} i \frac{\mu_0 \omega_0^2}{2k_0} \hat{T}^2 \mathcal{P}_{\text{NL}} &= i \frac{\mu_0 \omega_0^2}{2k_0} \varepsilon_0^2 n_0^2 c n_2 \hat{T}^2 [|\mathcal{E}|^2 \mathcal{E}] \\ &= i \frac{\omega_0^2 \varepsilon_0 n_0^2 n_2}{2k_0 c} \hat{T}^2 [|\mathcal{E}|^2 \mathcal{E}] \quad (\mu_0 \varepsilon_0^2 = \varepsilon_0 / c^2) \\ &= i \frac{\omega_0}{c} n_2 \hat{T}^2 [I \mathcal{E}] \quad (k_0 = n_0 \omega_0 / c, \quad \varepsilon_0 n_0 |\mathcal{E}|^2 = 2I / c), \end{aligned} \quad (21)$$

that is,

$$\boxed{\text{Kerr} = i K_0 n_2 \hat{T}^2 [I \mathcal{E}], \quad K_0 \equiv \frac{\omega_0}{c} \text{ (vacuum wavenumber).}} \quad (22)$$

This is *identical* to the Gaussian result. The agreement is a genuine cross-check: the 4π 's and the ε_0 's cancel because n_2 is defined through the intensity in both systems.

6 Self-steepening

This section is self-contained. It replaces the qualitative paragraph “the more intense peak travels more slowly than the wings” by an explicit derivation, and identifies exactly which part of that statement is right and which part is hand-waving.

6.1 Extracting the shock term

Start from the Kerr term (22) after division by $\hat{U}$, which by (16) leaves a single $\hat{T}$. Keep only the Kerr nonlinearity (no diffraction, no dispersion, no plasma) so that the effect is isolated:

$$\frac{\partial \mathcal{E}}{\partial z} = i K_0 n_2 \hat{T} [I\mathcal{E}] = \underbrace{i K_0 n_2 I \mathcal{E}}_{\text{SPM}} - \underbrace{\frac{n_2}{c} \frac{\partial}{\partial \tau} (I\mathcal{E})}_{\text{shock}}, \quad (23)$$

where the second term used $\hat{T} = 1 + \frac{i}{\omega_0} \partial_\tau$ and $K_0/\omega_0 = 1/c$. Define the shock coefficient

$$s \equiv \frac{n_2}{c}. \quad (24)$$

(If one keeps $\hat{T}^2$ without dividing by $\hat{U}$, the coefficient is $2s$; this is the origin of the $2n_2/c$ quoted in the report. The physically meaningful value, after $\hat{U}$ has been divided out, is s .)

6.2 The intensity obeys a Burgers equation

Write the envelope in amplitude–phase form, $\mathcal{E} = \sqrt{I} e^{i\varphi}$ (absorbing the constant of (2), which does not affect the structure). Then

$$\frac{\partial \mathcal{E}}{\partial z} = \left[\partial_z \sqrt{I} + i \sqrt{I} \partial_z \varphi \right] e^{i\varphi}, \quad \frac{\partial}{\partial \tau} (I\mathcal{E}) = \left[\partial_\tau (I^{3/2}) + i I^{3/2} \partial_\tau \varphi \right] e^{i\varphi}.$$

Separating real and imaginary parts of (23):

$$(\text{real}) \quad \partial_z \sqrt{I} = -s \partial_\tau (I^{3/2}) = -\frac{3}{2} s \sqrt{I} \partial_\tau I, \quad (25)$$

$$(\text{imag}) \quad \partial_z \varphi = K_0 n_2 I - s I \partial_\tau \varphi. \quad (26)$$

Multiplying the real equation by $2\sqrt{I}$ gives the central result:

$$\boxed{\frac{\partial I}{\partial z} + 3s I \frac{\partial I}{\partial \tau} = 0, \quad s = \frac{n_2}{c}.} \quad (27)$$

This is the **inviscid Burgers equation**, the archetype of shock formation. It is what justifies the word “steepening”, and everything qualitative about self-steepening follows from it without further argument:

- (i) **Each intensity level travels at its own speed.** Along the characteristics $d\tau/dz = 3sI$, the intensity is constant. High-intensity points drift toward *larger* τ . Since $\tau = t - z/v_g$, larger τ means arriving later: the peak is retarded relative to the wings. The statement “the peak travels more slowly” is therefore correct.
- (ii) **The distortion is intrinsically asymmetric.** The advection velocity depends on I , not on $\partial_\tau I$. The peak overtakes the trailing edge (which lies at larger τ) and runs away from the leading edge. The trailing edge compresses and steepens; the leading edge flattens. A perfectly symmetric Gaussian input therefore becomes asymmetric immediately, which is the point the qualitative paragraph was trying to make.
- (iii) **A shock forms at finite distance** (next subsection).

6.3 Shock distance

For $\partial_z I + c(I)\partial_\tau I = 0$ with $c(I) = 3sI$, characteristics cross, and $\partial_\tau I$ becomes infinite, at

$$z_{\text{sh}} = \frac{-1}{\min_\tau \frac{\partial c(I_0)}{\partial \tau}} = \frac{-1}{3s \min_\tau \partial_\tau I_0}. \quad (28)$$

For a Gaussian input $I_0(\tau) = I_p \exp(-\tau^2/t_p^2)$, the steepest negative slope sits at $\tau = t_p/\sqrt{2}$ and equals $\min_\tau \partial_\tau I_0 = -\sqrt{2} e^{-1/2} I_p/t_p \simeq -0.858 I_p/t_p$. Hence

$$z_{\text{sh}} = \frac{1}{3\sqrt{2}e^{-1/2}} \frac{ct_p}{n_2 I_p} \simeq 0.39 \frac{ct_p}{n_2 I_p}. \quad (29)$$

The numerical coefficient 0.39 reproduces the classical result of Anderson and Lisak, quoted by Agrawal as $z_{\text{sh}} = 0.39 L_{\text{NL}}/s'$ with $L_{\text{NL}} = 1/(K_0 n_2 I_p)$ and $s' = 1/(\omega_0 t_p)$. Recovering it from the Burgers picture is a useful check that the factor 3 in (27) is right.

How far is that, in practice? Comparing to the self-focusing length $L_{\text{SF}} = 1/(n_2 k_0 I_p)$ of the same beam,

$$\frac{z_{\text{sh}}}{L_{\text{SF}}} = 0.39 n_0 \omega_0 t_p. \quad (30)$$

For fused silica at 800 nm with $t_p = 136$ fs, $\omega_0 t_p \simeq 320$ and $n_0 = 1.4533$, giving $z_{\text{sh}} \simeq 180 L_{\text{SF}}$. Self-steepening is thus a very slow effect compared with self-focusing: on its own it would never matter. It becomes visible in filamentation only because the collapse first compresses the pulse in time (small t_p) and raises the intensity (large I_p), both of which shorten z_{sh} dramatically. This is why the shock term is negligible on the leading edge of the propagation and abruptly important near the nonlinear focus.

6.4 Spectral signature: why the blue side runs away

From the imaginary part, to leading order the phase is the ordinary SPM phase $\varphi \simeq K_0 n_2 I z$, so the instantaneous frequency shift is

$$\delta\omega(\tau) = -\frac{\partial \varphi}{\partial \tau} = -K_0 n_2 z \frac{\partial I}{\partial \tau}. \quad (31)$$

On the leading edge $\partial_\tau I > 0$, so $\delta\omega < 0$: red shift. On the trailing edge $\partial_\tau I < 0$, so $\delta\omega > 0$: blue shift. Without the shock term the two edges are mirror images and the spectrum is symmetric.

With the shock term, (27) makes $|\partial_\tau I|$ on the trailing edge grow without bound as $z \rightarrow z_{\text{sh}}$, while the leading edge flattens. Through (31) the blue edge of the spectrum therefore extends much further than the red edge, and diverges at the shock. This is the mechanism behind the blue-shifted asymmetry of supercontinuum spectra [1, 2], and it is a *prediction* of (27) rather than an assertion.

6.5 What the “intensity-dependent group velocity” argument gets wrong

The usual one-line justification is: the nonlinear index adds $n_2 I$ to the group index, so $1/v_g \rightarrow (n_g + n_2 I)/c$, and in the retarded frame a point of intensity I drifts at

$$\left. \frac{d\tau}{dz} \right|_{\text{naive}} = \frac{n_2 I}{c} = s I. \quad (32)$$

Compare with the exact characteristic velocity of (27), $d\tau/dz = 3sI$. The naive argument gets the *sign* right (peak retarded) and the *scaling* right (linear in I), but underestimates the drift

by a factor 3. The missing factor comes from the shock operator acting on the product $I\mathcal{E}$, not on the phase alone: differentiating $I^{3/2}$ produces $\frac{3}{2}I^{1/2}\partial_\tau I$, and converting back from $\sqrt{I}$ to I contributes the remaining factor 2.

So the qualitative picture in the report is not wrong, but it should not be presented as a derivation. The honest formulation is: *the shock term makes the intensity obey a Burgers equation whose characteristic velocity is $3n_2I/c$; an intensity-dependent group velocity is a useful mental image for the sign of the effect, but it does not give the correct coefficient.*

Check. The report currently writes the shock term as $(2n_2/c)\partial(|\mathcal{E}|^2\mathcal{E})/\partial t$. The factor 2 is consistent with keeping $\hat{T}^2$ and using K_0 rather than k_0 , but the quantity being differentiated should be the *intensity* times the envelope, $I\mathcal{E}$, not $|\mathcal{E}|^2\mathcal{E}$ – the two differ by $\frac{1}{2}\varepsilon_0 n_0 c$. Worth fixing when the paragraph is rewritten.

6.6 Numerical verification

Everything above is checkable without the full solver, since (23) is a one-dimensional equation. The module `sim/self_steepening.py` integrates it with RK4 and a spectral ∂_τ , and produces two figures.

The Burgers law, tested rather than asserted. Differentiating (27) along a characteristic gives $dJ/dz = -3sJ^2$ for $J = \partial_\tau I$, hence

$$\frac{|J|(z)}{|J|(0)} = \frac{1}{1 - z/z_{\text{sh}}}. \quad (33)$$

A threshold at G times the initial slope must therefore be crossed at exactly $z = (1 - 1/G)z_{\text{sh}}$, independently of t_p , I_p and n_2 . Measuring that crossing for $G = 3$ over input durations $t_p = 15$ to 80 fs gives $z_{\text{meas}}/z_{\text{sh}} = 0.6667$ with a maximum deviation of 1×10^{-4} from the predicted $2/3$, and the slope-growth curves for all six durations collapse onto (33). This validates the factor 3 in (27) and the coefficient 0.39 in (29) simultaneously.

The spectral asymmetry. Propagating a 30 fs, 5×10^{13} W/cm² pulse to $0.85 z_{\text{sh}}$:

	shock OFF	shock ON
trailing-edge slope $\partial_\tau I$ (norm.)	−0.0286	−0.1901
leading-edge slope $\partial_\tau I$ (norm.)	+0.0286	+0.0155
instantaneous frequency range (rad/fs)	[−0.67, +0.67]	[−0.43, +1.62]
spectral extent, blue/red at 10^{-3}	1.00	3.68

Without the shock term the pulse shape is rigorously unchanged (SPM is a pure phase) and every quantity is symmetric to machine precision. With it, the trailing edge steepens by $6.6\times$ while the leading edge flattens, exactly as (27) requires, and the blue edge of the spectrum runs $3.7\times$ further than the red one.

A caution on the metric. The natural-looking measure – the ratio of spectral *energy* above and below ω_0 – is the wrong observable here: it comes out slightly *red*-heavy (0.75) even in the case where the blue edge extends four times further. The reason is that the blue tail is generated by a single narrow region of τ (the steepened edge), so it is spread thinly over a wide bandwidth, whereas the bulk of the energy stays in the central SPM peaks. The asymmetry claimed here is one of spectral *extent*, not of energy, and the figure reports the extent.

7 Delayed Raman response

The Raman derivation is unit-independent once the driving term is normalized, so it carries over unchanged. A Raman-active vibrational mode of coordinate q , frequency Ω_R and damping time τ_d is driven by the cycle-averaged field through the Placzek polarizability expansion $\alpha(q) \simeq \alpha_0 + \alpha' q$:

$$\mu \left[\frac{d^2}{dt^2} + \frac{2}{\tau_d} \frac{d}{dt} + \Omega_R^2 \right] q = \frac{1}{2} \alpha' \langle \mathbf{E}^2 \rangle = \frac{1}{4} \alpha' |\mathcal{E}|^2. \quad (34)$$

Normalizing to $Q \equiv q \Omega_R^2 / \kappa$ with $\kappa = \alpha' / (4\mu)$, and expressing the drive as an intensity,

$$\left[\frac{d^2}{dt^2} + \frac{2}{\tau_d} \frac{d}{dt} + \Omega_R^2 \right] Q = \Omega_R^2 I. \quad (35)$$

The Green function of this operator, causal and normalized by the jump condition $G'(0^+) - G'(0^-) = 1$, is

$$G(t) = \tau_s e^{-t/\tau_d} \sin(t/\tau_s) \mathcal{H}(t), \quad \frac{1}{\tau_s} = \sqrt{\Omega_R^2 - \frac{1}{\tau_d^2}}, \quad (36)$$

so the response function is

$$R(t) = \Omega_R^2 \tau_s e^{-t/\tau_d} \sin(t/\tau_s) = \frac{\tau_s^2 + \tau_d^2}{\tau_s \tau_d^2} e^{-t/\tau_d} \sin(t/\tau_s), \quad (37)$$

using $\Omega_R^2 = \tau_s^{-2} + \tau_d^{-2}$. The second form is the one usually quoted in the filamentation literature. It is correctly normalized:

$$\int_0^\infty R(t) dt = \Omega_R^2 \tau_s \cdot \frac{1/\tau_s}{\tau_d^{-2} + \tau_s^{-2}} = 1. \quad (38)$$

The Kerr response is then split between an instantaneous and a delayed part with weight f_R :

$$I \longrightarrow (1 - f_R) I(\mathbf{r}, t) + f_R \int_{-\infty}^t R(t - \tau) I(\mathbf{r}, \tau) d\tau. \quad (39)$$

The solver uses $f_R = 0.18$, $\tau_d = 32$ fs, $\tau_s = 12$ fs.

Check. In the Gaussian document the normalized equation is written once with $\Omega^2 |\mathcal{E}|^2$ on the right and once with $\Omega^2 I$; these differ by $n_0 c / 8\pi$. Only the second is consistent with R being normalized to unit area and with I entering (39). The Q substitution is also performed twice in a row.

8 Plasma response: the Drude model

A conduction electron of effective mass m_{eff} , driven by the field and damped by collisions at rate $1/\tau_c$:

$$m_{\text{eff}} \frac{d\mathbf{v}}{dt} = -e\mathbf{E} - \frac{m_{\text{eff}}}{\tau_c} \mathbf{v}. \quad (40)$$

Seeking $\mathbf{v} = \frac{1}{2} \tilde{\mathbf{v}} e^{i\psi} + \text{c.c.}$ and using the SVEA ($|\partial_t \tilde{\mathbf{v}}| \ll \omega_0 |\tilde{\mathbf{v}}|$, so $d\mathbf{v}/dt \simeq -i\omega_0 \tilde{\mathbf{v}}$):

$$\tilde{\mathbf{v}} = -\frac{e\tau_c}{m_{\text{eff}}(1 - i\omega_0\tau_c)} \mathcal{E} \implies \mathcal{J}_{\text{pl}} = -e\rho \tilde{\mathbf{v}} = \frac{e^2 \rho \tau_c (1 + i\omega_0\tau_c)}{m_{\text{eff}}(1 + \omega_0^2 \tau_c^2)} \mathcal{E}. \quad (41)$$

The corresponding source term in the propagation equation is

$$\begin{aligned} \frac{\mu_0}{2ik_0} \frac{\partial \mathcal{J}_{\text{pl}}}{\partial t} &\simeq \frac{\mu_0}{2ik_0} (-i\omega_0) \mathcal{J}_{\text{pl}} = -\frac{\mu_0 \omega_0}{2k_0} \frac{e^2 \tau_c}{m_{\text{eff}}(1 + \omega_0^2 \tau_c^2)} \rho (1 + i\omega_0\tau_c) \mathcal{E} \\ &= -\frac{e^2 \omega_0 \tau_c}{2k_0 \varepsilon_0 c^2 m_{\text{eff}}(1 + \omega_0^2 \tau_c^2)} \rho (1 + i\omega_0\tau_c) \mathcal{E}. \end{aligned} \quad (42)$$

Defining the inverse-bremsstrahlung cross section in SI,

$$\sigma = \frac{k_0 e^2}{n_0^2 \omega_0^2 \varepsilon_0 m_{\text{eff}}} \cdot \frac{\omega_0 \tau_c}{1 + \omega_0^2 \tau_c^2}, \quad (43)$$

and using $n_0^2 \omega_0^2 = k_0^2 c^2$, the prefactor above is exactly $\sigma/2$:

$$\frac{\sigma}{2} = \frac{e^2 \omega_0 \tau_c}{2 k_0 c^2 \varepsilon_0 m_{\text{eff}} (1 + \omega_0^2 \tau_c^2)}. \quad (44)$$

Hence

$$\text{plasma} = -\frac{\sigma}{2} (1 + i \omega_0 \tau_c) \rho \mathcal{E}, \quad (45)$$

the real part being absorption (inverse bremsstrahlung) and the imaginary part the defocusing phase. Expression (43) is exactly what the solver computes:

$$\text{sigmaomega} = k_0 * q_e ** 2 * \tau_c / (n_0 ** 2 * m * \text{eps0} * \omega_0 * (1 + (\omega_0 * \tau_c) ** 2))$$

so this equation is the direct justification of that line of code. No Gaussian detour is needed.

9 Photoionization losses

Energy conservation (Poynting) requires that the power transferred to the newly freed electrons equals the gap energy times the ionization rate:

$$\frac{\partial u}{\partial t} + \nabla \cdot \mathbf{S} = -\mathbf{J}_{\text{ion}} \cdot \mathbf{E}, \quad \langle \mathbf{J}_{\text{ion}} \cdot \mathbf{E} \rangle_T = U_i W_{\text{PI}}. \quad (46)$$

Taking the current in phase with the field, $\mathbf{J}_{\text{ion}} = \gamma_{\text{ion}} \mathbf{E}$, and using $\langle \mathbf{E}^2 \rangle = \frac{1}{2} |\mathcal{E}|^2$:

$$\gamma_{\text{ion}} \frac{1}{2} |\mathcal{E}|^2 = U_i W_{\text{PI}} \implies \gamma_{\text{ion}} = \frac{2 U_i W_{\text{PI}}}{|\mathcal{E}|^2}. \quad (47)$$

The source term follows the same route as the plasma one, keeping a single $\hat{T}$ because only one time derivative acts on $\mathbf{J}$:

$$\begin{aligned} -\frac{\mu_0 \omega_0}{2 k_0} \hat{T} [\gamma_{\text{ion}} \mathcal{E}] &= -\frac{\mu_0 \omega_0}{2 k_0} \hat{T} \left[\frac{2 U_i W_{\text{PI}}}{|\mathcal{E}|^2} \mathcal{E} \right] \\ &= -\frac{\mu_0 \omega_0}{2 k_0} \varepsilon_0 n_0 c \hat{T} \left[\frac{U_i W_{\text{PI}}}{I} \mathcal{E} \right] \quad (|\mathcal{E}|^2 = 2 I / \varepsilon_0 n_0 c) \\ &= -\frac{1}{2} \hat{T} \left[\frac{U_i W_{\text{PI}}}{I} \mathcal{E} \right] \quad (\mu_0 \varepsilon_0 n_0 c \omega_0 / k_0 = 1). \end{aligned} \quad (48)$$

The factor $\frac{1}{2}$ therefore comes out identical to the Gaussian derivation, confirming that it is not a units artifact. Dimensionally $[U_i W_{\text{PI}} / I] = \text{m}^{-1}$, as an absorption coefficient must be: the denominator is an *intensity*, not a squared field amplitude.

The rate W_{PI} is the Keldysh photoionization rate,

$$W_{\text{PI}} = \frac{2 \omega_0}{9 \pi} \left(\frac{\omega_0 m}{\hbar \sqrt{\Gamma_1}} \right)^{3/2} \left[\sqrt{\frac{\pi}{2 K(\Gamma_2)}} \sum_{n=0}^{\infty} e^{-n \alpha} \Phi \left(\sqrt{\beta(n+2\nu)} \right) \right] e^{-\alpha \langle x+1 \rangle}, \quad (49)$$

with $\gamma = \omega_0 \sqrt{m U_i} / (e |\mathcal{E}|)$, $\Gamma_1 = \gamma^2 / (1 + \gamma^2)$, $\Gamma_2 = 1 / (1 + \gamma^2)$, $\alpha = \pi [K(\Gamma_1) - E(\Gamma_1)] / E(\Gamma_2)$, $\beta = \pi^2 / [4 K(\Gamma_2) E(\Gamma_2)]$, $x = (2 U_i / \pi \hbar \omega_0) E(\Gamma_2) / \sqrt{\Gamma_1}$ and $\nu = \langle x+1 \rangle - x$; K and E are the complete elliptic integrals (argument $m = k^2$) and Φ the Dawson function. The limits $\gamma \gg 1$ and $\gamma \ll 1$ give multiphoton and tunnel ionization respectively.

10 The complete equation

Collecting (22), (39), (45) and the ionization term:

$$\boxed{\begin{aligned} \hat{U} \frac{\partial \mathcal{E}}{\partial z'} = & i \left[\frac{\nabla_{\perp}^2}{2k_0} + \hat{L} \left(\hat{U} + \frac{\hat{L}}{2k_0} \right) \right] \mathcal{E} \\ & + i K_0 n_2 \hat{T}^2 \left[(1 - f_R) I + f_R \int_{-\infty}^t R(t - \tau) I(\tau) d\tau \right] \mathcal{E} \\ & - \frac{1}{2} \hat{T} \left[\frac{U_i W_{\text{PI}}}{I} \right] \mathcal{E} - \frac{\sigma}{2} (1 + i\omega_0 \tau_c) \rho \mathcal{E}. \end{aligned}} \quad (50)$$

Structure. Terms carrying an explicit i are conservative and act as phases: diffraction, dispersion, Kerr, and plasma defocusing (the $i\omega_0 \tau_c$ part). Terms without i are dissipative: photoionization loss and inverse bremsstrahlung. This is worth stating because the compact notation $\partial_z \mathcal{E} = i\hat{N}\mathcal{E}$ found in the literature is not literally consistent term by term: with a global i pulled out, the Kerr contribution inside $\hat{N}$ is real while the loss terms must themselves carry an i , so that $i\hat{N}$ reproduces real losses. The boxed form (50) avoids that trap by keeping every factor explicit.

Agreement with the Gaussian derivation. Every physical prefactor came out the same: the Kerr coefficient $K_0 n_2$, the ionization factor $-\frac{1}{2}$, and the plasma factor $-\sigma/2$. That is expected — the two unit systems must agree — but it is a real check, because the intermediate expressions look nothing alike. The place where they genuinely differ is $\chi^{(3)}$, which is a different number in the two systems, and which is why n_2 (not $\chi^{(3)}$) should always be the quantity carried through the code.

A Gaussian $\leftrightarrow$ SI dictionary

Quantity	Gaussian	SI
Displacement field	$\mathbf{D} = \mathbf{E} + 4\pi\mathbf{P}$	$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P}$
Source prefactor	$4\pi/c^2$	$\mu_0 = 1/\varepsilon_0 c^2$
Intensity	$I = \frac{n_0 c}{8\pi} \mathcal{E} ^2$	$I = \frac{\varepsilon_0 n_0 c}{2} \mathcal{E} ^2$
Nonlinear index	$n_2 = \frac{12\pi^2 \chi^{(3)}}{n_0^2 c}$	$n_2 = \frac{3\chi^{(3)}}{4\varepsilon_0 n_0^2 c}$
Cross section	$\sigma = \frac{4\pi e^2}{k_0 c^2 m_{\text{eff}}} \frac{\omega_0 \tau_c}{1 + \omega_0^2 \tau_c^2}$	$\sigma = \frac{k_0 e^2}{n_0^2 \omega_0^2 \varepsilon_0 m_{\text{eff}}} \frac{\omega_0 \tau_c}{1 + \omega_0^2 \tau_c^2}$
Kerr term	$i K_0 n_2 \hat{T}^2 [I\mathcal{E}]$	$i K_0 n_2 \hat{T}^2 [I\mathcal{E}]$ (identical)
Ionization term	$-\frac{1}{2} \hat{T} [U_i W_{\text{PI}}/I] \mathcal{E}$	identical
Plasma term	$-\frac{\sigma}{2} (1 + i\omega_0 \tau_c) \rho \mathcal{E}$	identical

B Points to check in the Gaussian document

Collected here for convenience; each is also flagged in place above.

1. **The section “The Lorentz model for STE” is a verbatim copy of the Drude section.** Everything after the hydrogenic-exciton introduction repeats the free-electron derivation word for word, including the sentence “The density of current created by a plasma of

density ρ'' and the final boxed equation. The Lorentz-oscillator derivation itself (bound resonance at ω_{tr} , giving $\Delta n \propto +\rho_s/(\omega_{tr}^2 - \omega^2)$ with a *positive* sign, opposite to Drude) is missing. This is the one substantive gap rather than a typo.

2. **J_{STE} appears in the wave equation but not in Ampère’s law**, where only $\mathbf{J}_{\text{ions}}$ and $\mathbf{J}_{\text{plasma}}$ are listed.
3. $\nabla \times \tilde{\mathbf{H}} = \nabla \times \tilde{\mathbf{B}}$ should be flagged as assuming $\mu = 1$.
4. **Raman drive written twice**, once as $\Omega^2|\mathcal{E}|^2$ and once as $\Omega^2 I$; the Q substitution is also duplicated.
5. **Poynting theorem** written $\partial_t u + \nabla S$; should be $\nabla \cdot \mathbf{S}$.
6. **Section heading “Comparison with”** is incomplete.

References

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